

Agile Software Factory for Student Service Learning

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Abstract

Service-learning is a teaching and learning strategy that integrates meaningful community service with instruction and reflection to enrich the learning experience, teach civic responsibility, and strengthen communities. By incorporating service-learning in Software Engineering classes, the student experience can be enhanced. Our approach was to create the Agile Software Factory within the department of Computer Science to handle the administrative tasks associated with a quality classroom service-learning experience. This paper describes the mission and the objectives of the Factory.

1. Introduction

It has been well-documented and understood that students learn the craft of software development better by working on real-world projects with real customers. For at least twenty years, our software development courses in the department of Computer Science (CS) have occasionally provided services to real-world clients. The department has also frequently received software requests from community members who have needs but cannot afford the high cost of software development. However, it was only occasional that the department could act on these requests as there was no central authority that could manage the requests and route them to channels in which the software need could be filled.

The Agile Software Factory (Factory) was created to fulfill the need in the department to have a community project clearinghouse, to manage relationships with existing community partners while soliciting new community partners and projects, and to provide facilities and knowledge to promote service-learning for students who are pursuing a Computer Science degree. As defined in the *Faculty Toolkit for Service-Learning in Higher Education* [1], service-learning is a teaching and learning strategy that integrates meaningful community service with instruction and reflection to enrich the learning experience, teach civic responsibility, and strengthen communities.

By using a service-learning approach students are not only able to learn important software engineering skills in a structured classroom and laboratory environment but they are also able to provide high-quality software to organizations that would be otherwise unable to meet their software requirements. Students are also exposed to the real-world problems associated with software engineering and gain valuable experiences in situations that they will be required to face in industry.

Already there are a number of other computer science departments using a similar service-learning approach teaching software engineering through service-learning [3, 4, 5, 6, 7]. The Agile Software Factory was patterned after these other programs, drawing from their experiences and combining it with our own unique history and academic culture.

Two challenges come up in a service-learning approach to software development courses. One was that it takes a significant amount of time to find suitable software projects for the course. An ideal project should contain enough complexity for student learning, and its size should be reasonable so that student teams are capable of producing a quality system in a restricted timeframe.

The other challenge encountered was the potential software maintenance issues after the semester ended. Students are not responsible for maintaining the software systems after the end of the semester, but the systems are in production use and like all production software need periodic maintenance. Other faculty members in the CS department have also encountered the similar challenges when supervising student projects, group or individual, that involve real customers. The Factory solves the challenges it faces by (1) creating a repository for software requests from community partners, and (2) providing ongoing software services through various student service-learning arrangements.

The Factory is also involved with avocation of Agile methodologies both in the classroom and in the greater industry. When working on real-world projects with real customers, producing quality working systems play a major role in customer satisfaction. Agile methodologies promote iterative and incremental development, which makes producing a quality working system within a short period of time, such as a semester, highly probable. In this avocation role, the Factory acts as a store of knowledge for the student community, providing opportunities for students to learn about Agile methods both in and outside of the classroom. The Agile Software Factory also pursues research into both methods of education about Agile methodologies and ways to evaluate these methods.

The idea of a software factory to facilitate software engineering education is nothing new. Siqueira, Barbaran and Becerra [2] applied the concept of software factory to a software engineering laboratory to create a software development environment that is close to the real world. Their approach emphasized on well-defined processes in a controlled environment. Chase, Oakes and Ramsey [8] created a small project support center to ease the challenges of using live projects for student service learning. The listed challenges of using live projects, in addition to pedagogical demands on the instructor, include maintenance concern, project scope, student skill specialization, data confidentiality, project solicitation and consistency of the projects. While the idea of software factory or support center is not new, the literature in this area is surprisingly scarce.

2. Factory mission & overview

The Agile Software Factory was developed out a need for a persistent managerial structure to exist within the department of Computer Science to facilitate student service-learning. To achieve this goal, the Factory was tasked with the following mission: 1) to develop community partners and relationships as both a source of support and student projects and 2) to espouse software engineering research particularly in Agile methodology. The Agile Software Factory, in relation to its primary mission, is also responsible for supporting the larger needs of the department by showcasing both the work the department does in software engineering and through evaluation and publication of both Factory programs and student work.

According to a recent internal analysis of two alumni surveys by our department, the majority of CS graduates who enter the software engineering field develop software for private companies. The report stated that “By a wide margin, they choose careers in private industry in programming and software engineering careers.” Also in the same report, a survey of incoming

freshmen found that “in general, incoming freshmen are seeking career paths that match those taken by our alumni, which is in private industry with an interest in programming.” Including real-world projects in the curriculum will increase student marketability, create scholarship engagement opportunities, and support student retention and recruitment. The Factory will further the university mission not only by providing a premier learning environment for students, but also by engaging the community in a meaningful and scholarly fashion.

The Agile Software Factory is tasked then, with assisting the department in providing a valuable, real-world service-learning experience to students. The Factory will achieve its task by acting as manager and clearing-house for student service-learning projects and will pursue and develop strong relationships with community partners in a mutually beneficial manner. The Agile Software Factory will also research into and provide information about Agile methods of software engineering.

The Factory will also strive to become self-sufficient financially in order to further the mission of agile software research and development. To achieve this goal, it will: (1) solicit corporate sponsorships and/or memberships, (2) apply for internal and external grants, and (3) commercialize potential software systems to generate revenues.

3. Projects

To facilitate a good service-learning experience for students, it is important to find and have good projects available for students to work on. Because the objective of service-learning, it is important that these projects come from the community at large. Projects must also have some academic merit, since it will be used within a classroom setting. The ideal project should be large and complex enough that students find them challenge, and at the same time, flexible and scalable so that a quality product can be developed by a team of students.

Before creation of the Agile Software Factory, it was necessary for an instructor who wished to do a service-learning project to solicit a project from the wider community. Because no central clearing-house of projects existed to connect professors with needy community partners, it became an extra burden on the professor to develop the community resources to solicit and develop projects for students to work on. Because of the extra work-load many professors chose not to use the service-learning approach in their classes, even when it may be the most effective means of teaching the subject. With the Agile Software Factory acts as a central clearing-house of projects, a significant burden is lifted from anyone who wishes to use a service-learning approach in their classroom. This increases the usefulness of the service-learning approach since much of the management required to create the proper service-learning environment for the student is handled by a third-party, the Factory. It is anticipated that this will enable more service-learning within the department curriculum.

The Agile Software Factory solicits projects throughout the year from needy community partners and works with the project partners to get as much detail as possible about the project so instructors who wish to use a project in their class can make informed decisions about what projects would be best. The Factory also works with community partners to identify possible projects within the partner’s organization as a way to increase the number of potential project candidates and to assist willing community partners to further develop and improve their business processes. The full project lifecycle is illustrated in Figure 1.

After students are finished with their service-learning experience, whether for a class or as an independent project, they submit their work back to the Agile Software Factory. When

projects are submitted back to the Factory, they are evaluated to determine the quality and future of the project. The quality of our projects is of the utmost concern to the Factory. Because much of our strategy is built on maintaining good relationships with our community partners, we strive to ensure that we provide them with the best software possible.

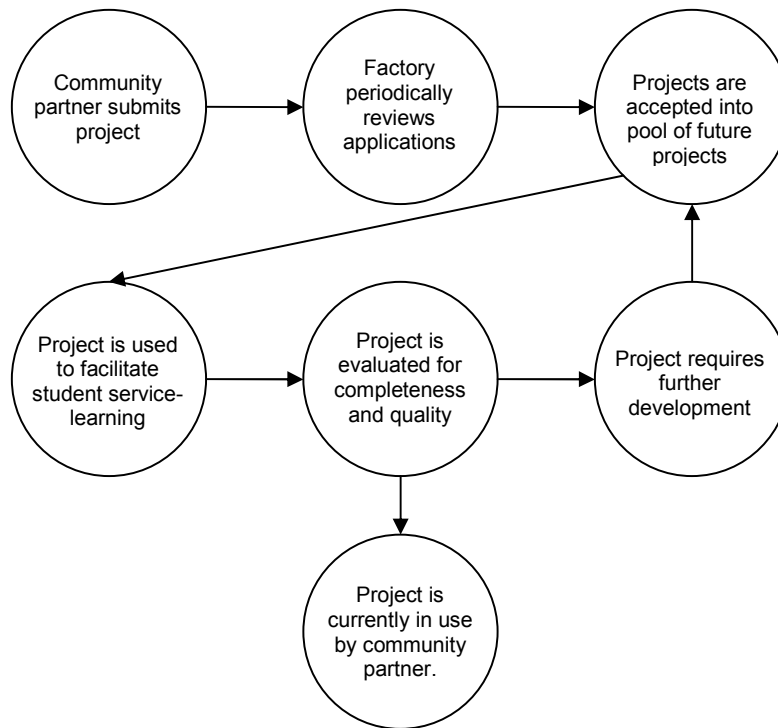


Figure 1. Factory Project Flow

The future of most projects that are handled through the Agile Software Factory is determined by the community partner who initiated the project. However, in rare instances, a project may have some commercial appeals and the Factory evaluates completed projects to determine if this is so. If a project could have future commercial use, it will be held in reserve and further developed by the Factory into a commercial project for the market at large.

As community partners deploy and use applications that students have developed in service-learning experiences, the community partner may want or need additional features or wish to further develop their project into a more useful tool. It may be inappropriate to use these project continuations in a classroom, so the Factory may employ students of various skill levels to do further work on projects providing further service-learning experiences for interested students and enhancing our support of the community. This offers the opportunities for student employment which furthers the Factory's mission of providing student service-learning.

At the time of writing this paper, the Factory has eight concurrent software development projects with eight community partners. Six of the projects are being used as software engineering service-learning course projects, one is under development by a graduate student as a graduation project, and the other project is being worked by a student employed by the Factory. The Factory has recently received a sponsorship from Agile Alliance [9], which will allow hiring more students to work on more projects in the near future.

4. Student service-learning

There are two approaches the Factory provide service-learning opportunities to students: 1) through service-learning courses and 2) through direct hiring of student employees.

Our approach to service-learning courses is like many other service-learning software engineering courses at other universities. We provide the service-learning experience through a controlled classroom and laboratory environment with instruction about the various software engineering tools and methodologies that are common-place in industry. These courses are typically taken by juniors, seniors as well as first-year graduate students who are interested in a career in software engineering. Students who desire to do service-learning in their Computer Science curriculum are enabled because of the Agile Software Factory. These opportunities generally take the form of a Software Engineering course offered within the department, but other options do exist both inside and outside of the classroom.

The Software Engineering course is structured within the department curriculum, with lectures, homework, and tests. However, a significant portion of class is devoted to the service-learning projects. The course content may vary among instructors but all major topics in software engineering are covered. When necessary, class time may be given for project team meetings and other project business as well as, to ease the scheduling burden on the students. Also there are lectures on typical software engineering tools like source-control and UML modeling software. The rest of the time is devoted to lectures on relevant topics to software engineering. Instructors of the Software Engineering class usually are mainly involved with the classroom work and are generally not involved with project details past the first few weeks. Instructor usually focuses on the initial planning stages and to make sure the project goes smoothly for the students, and give students the total control while monitoring the progress. Furthermore, instructors often participate in the first customer meetings to make sure that the students and community partners are on the right track.

Students working on the service-learning projects are provided with facilities in which to work on their project. They also usually are not required to setup or configure any technology they may be required to use in their project. The setup of technology is the responsibility of the Factory and is done in the lead-up to the start of a project.

Another approach to service-learning provided by the Factory is through part-time employment of students. Depending on the funding situation, a small handful of students may elect to work directly with the Factory on projects that the Factory needs or wants to further develop. These may be projects that were not worked on in the Software Engineering class due to time or resource constraints, they may be past projects that are in need of further development by community partners, or it may be commercialization of past projects. Students who chose this path are usually committed to the project for a fixed number of hours a week and generally get financial assistance through the department or a relevant community partner, as an assistantship or other graduate student stipend. The Factory is also actively applying for funding in order to provide part-time student employees.

The results and responses given by students who complete a service-learning experience are similar to those responses from other universities. Students were both pleased with the experience provided and were glad to have participated in the project. The students are also happy to get hands-on experience learning about what software engineering is and what a possible career will be like when the graduate.

5. Community involvement and interaction

Community involvement is an integral part of the Agile Software Factory. There are two ways for community partners to be involved with the Factory. Most of our community partners are sources of projects. Primarily they tend to be non-profit organizations that want or need software but don't have the budgets for professionally developed systems. In many cases, there is no commercial, off-the-shelf software available that meets the organization needs. The Factory helps these organizations develop the software they need, at a professional level of quality, for no cost. The other way for community partners to be involved with the Factory is through donations of time, money, and expertise. Without this type of support the scope of the Factory would be greatly limited.

Community partners that are actively working on projects with the Agile Software Factory are also working directly with the students. Since the Factory primarily focuses on the education of Agile methodologies, it is important that community partners also participate in the project development closely. Generally community partners can participate by providing customer information to student project groups on a real-time basis. By having the community partners directly in contact with the students, the student experience is enhanced with real-world, hands-on customer experience. One thing students learn in their service-learning project that they don't usually learn in other courses is the teamwork and communication skills. The direct contact also allows increase the student's ability to provide quality software through fast software debugging and validation testing ensuring the quality of the software products created by the students.

6. Long-term vision and future work

Flexibility is a key to the long-term health of the Agile Software Factory. Like a software development project, it is important for the Factory to quickly adapt to changes in the requirements responding to customer needs and complaints in an efficient manner. By changing with the needs of the department of Computer Science and our community partners the Agile Software Factory can best provide the quality service-learning experience that students and faculty desire.

For future improvement of the Factory, assessments are important and necessary. The Factory will perform assessment regularly to ensure the success of all Factory activities. Possible measurements may include:

- number of inquiries and software requests from community
- number of projects in-progress and completed
- number of systems in-use by the community partners
- customer satisfaction.

The Factory will also prepare instruments for measuring the outcome of student service-learning. The assessment activities may include:

- Observe student project meetings and customer meetings
- Prepare surveys for both students and their clients
- Interview students and customers
- Analyze project artifacts and software products

In the future, the Factory wishes to expand both the service-learning program, applying it to a wider range of software engineering classes in the department of CS. The Factory will also

expand its ability for long-term software maintenance of its projects that are currently deployed by community partners. Currently this is limited by available manpower and funds, but through the judicious use of grant funding it may be possible for the Factory to provide both long-term software support services to our community partners and an independent student-learning experience to interested students.

Other work that the Agile Software Factory will undertake is a program to further expand exposure of Agile methodologies to students and the wider community. We are deeply invested Agile methods, and believe they are both applicable and useful in the greater community. Not only will the Factory provide information and lectures on methods, but it will also perform research into methods of teaching the Agile methodology in the classroom including evaluation of the effectiveness of courses and students.

Over time we hope that the Agile Software Factory becomes an integral part of a student's software engineering education and a valued community partner which is a source of high-quality software. By developing good relationships with the community we provide a valuable service-learning experience that is used to give students a hands-on education in software engineering. By giving students a hands-on software engineering education we provide them with the tools and knowledge that they will need when they move on to industry.

7. Acknowledgements

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